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E-471 DISPATCH SYSTEM INTERFACES

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of the Dispatch System in accordance with Part 4, Subsection 1.2 Summary of Work.

The substation extension will be connected to the PLN system via the communication network. The interface will be for the following functions:

- (1) To connect the substations to a system wide voice communication network.
- (2) To permit control of the power plant and substation equipment from the Java Control Center (JCC) and to provide power plant and substation indications and measurements necessary for the system control.

2. CODES AND STANDARDS

The Contractor shall comply with the requirement of Part 4 Section 2.5 Codes and Standards for the responsibilities under this Contract.

2.1 Indonesian National Standards

SPLN : Standard Perusahaan Umum Listrik Negara.

DITJEN LPE : Direktorat Jendral Listrik dan Pemanfaatan Energi mengenai :
Aturan Jaringan Jawa-Madura-Bali

2.2 International Electrotechnical Commission

IEC 60870 : Telecontrol equipment and system

3. INTERFACE

3.1 General

The Contractor shall provide all connections between protection equipment, metering and measuring system, etc. to substation interface panel Block 2, 3 and 4 for distribution of the data to the SCADA system and others. Any interfacing equipment required to ensure a proper operation of the Contractor's relaying system shall be the Contractor's responsibility.

3.2 Relaying Interfaces

The Contractor shall provide the design of relaying interfaces as required.

For design purposes the signaling equipment shall be assumed to have contact-in to contact-out operating times as follows:

- (1) Permissive transfer trip : not exceeding 20 ms.
- (2) Blocking signal : operating 10 ms and reset time 10 ms.

The output contact ratings shall be assumed to be 2 ampere, at 220 V DC.

Contact inputs supplied by the Contractor shall be “voltage-free” and shall be rated for 2 amperes inductive at 220V DC.

3.3 Voice Communication

The communication system serves as a system wide internal communications network. The Contractor shall have provisions for interconnecting the power plant PABX system with the existing carrier system. The Contractor shall provide interconnection with the existing carrier equipment located elsewhere in the substation.

The Contractor shall provide special terminal jacks for use of portable maintenance hand set telephone mounted on each relay cabinet. These jacks shall be wired by the Contractor, as instructed by the Owner, to the fiber optic panels, using the channel for emergency and maintenance communication.

4. DISPATCH SYSTEM INTERFACES

4.1 General

The Java Control Center requires access to measuring quantities, control and position indications of circuit breakers and disconnecting switches, etc. These quantities are as a guide for the Contractor in designing the substation.

All the above-mentioned indications and signals from the substation and the new power plant, shall be collected in the substation interface panels.

The data required for the SCADA system shall be sent to the new SCADA interface panels, to the new ITF panel and further to the new RTU.

The necessary interconnections from the new RTU supplied by the Contractor to the existing PLC cabinet shall be provided. The new RTU protocol shall be able to communicate with the existing protocol of type HNZ.

All the above-mentioned equipment shall be placed in the substation control building. Refer to the drawing no. 1222-00-E-52-AB-006/007.

The designing and supply of all the above-mentioned panels and equipment is the responsibility of the Contractor.

A new Distributed Control System (DCS) shall be provided as the substation automatic control and monitoring system for interfacing between substation, SCADA and power plant, VDU, placed in the substation control room.

The VDU and printer shall be provided and placed in the substation control room for the purpose of controlling and monitoring the extension switchyard. This VDU shall always be able to execute all operations despite any possible problem in any other system.

Interface connection to the power plant central control room (dual redundancy plant bus) via fiber optic cable shall be executed under the power-plant's I & C Technical Specification I-005 and bid drawings.

4.2 Measuring Interfaces

The Contractor shall provide all transducers which requires for measuring indications.

The Contractor shall provide all signals as requested by the Owner's SCADA requirements as defined in "BUKU POLA SCADA" or "SCADA PATTERN BOOK". The signals shall be taken directly from CT and VT (Original Signal). As a minimum, the following measuring circuits are anticipated:

- (1) All Watts, VAR, and phase current connections.
- (2) All bus voltages.
- (3) Generator transformer voltages at EHV side.
- (4) Other signals to be defined in "BUKU POLA SCADA" to be given after the Contract award (for reference, the number is approximately 20).

4.3 Contact Inputs

The Contractor shall provide all necessary contact inputs as requested by the Owner. Contacts shall be suitable for 220 V DC.

The Contractor shall prepare all cabling including termination of these inputs to substation interface panel.

As a minimum, the following contact inputs are anticipated:

- (1) All bus bar circuit breaker positions (separate open and closed).
- (2) All disconnecting switch positions (separate open and closed).
- (3) Six (6) miscellaneous indications (two (2) separate inputs each).
- (4) Twenty (20) miscellaneous indications (one (1) input each).
- (5) Six (6) grouped alarm inputs of various functions as defined in “BUKU POLA SCADA” to be given later after the Contract award (including contacts in Clause 4.4 Paragraph 3).

4.4 Control Outputs

The Contractor shall design the control circuits of the devices to be controlled from the JCC. The list of devices to be controlled from the dispatch center will be as defined in “BUKU POLA SCADA”.

The Contractor’s circuit design provisions shall be such that the dispatch controls shall be in parallel with any other controls and shall not hinder the other control systems or negate any protective or other interlocking required for safe operation of the equipment.

The Contractor shall provide key locked “in-out” selector switches between the RTU I/O contacts and the controlled equipment. A contact from each selector switch shall be provided to the I/O cabinet for indication that the JCC does not have control of selected equipment.

5. PERFORMANCE TESTING AND GUARANTEES

The dispatch system furnished under this Specification shall be designed and arranged to facilitate periodically testing, calibration and maintenance.

Component, wiring and other generic panel testing requirements are covered in Technical Specifications I-002 and I-003.

The control system shall be assembled, and all hardware’s inputs and outputs shall be tested successfully in accordance with the Specification to the satisfaction of

the Owner, and all modifications requested by the Owner shall be implemented without any cost to the Owner.

All components after assembly shall be energized and “burned in” for a minimum of 90 days to minimize “infant mortality” problems with components.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control for the dispatch system shall conform to the requirements included in Part 4, Section 2, Subsection 2.7.

The quality assurance/quality control procedure specified shall include, but not be limited to, the followings:

- (1) Continuity and Wiring Tests
- (2) Insulation and High Potential Testing
- (3) Packaging and Shipping
- (4) Welding
- (5) Cleaning and Painting

The quality assurance/quality control documentation as specified above shall include, but not be limited to, the followings:

- (1) Shop Test Reports
- (2) All other documentation required by application codes and standards

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